JIBAR 3-Month Rates: Analysis, Modelling and Risk Implications

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BSc Applied Mathematics and Statistics

September 2026

Contents

Abstract

The Johannesburg Interbank Average Rate (JIBAR), particularly the 3-month tenor, has historically served as South Africa's primary benchmark interest rate for pricing loans, derivatives, and financial contracts. This study analyses the historical behaviour of 3-month JIBAR from 2011 to 2026 using monthly data obtained from FRED and the South African Reserve Bank.

Using descriptive statistics, time series visualisation, correlation analysis, and volatility measures, the research examines the relationship between JIBAR and the South African Repo Rate. The study further explores the implications of the ongoing global transition from Interbank Offered Rates (IBORs) to Risk-Free Rates (RFRs), with specific focus on South Africa's shift toward ZARONIA.

Key findings reveal significant volatility in JIBAR, especially during the COVID-19 pandemic and the 2022 global inflation surge, along with a strong positive correlation with the Repo Rate. The analysis highlights the increasing basis risk and transition challenges facing financial institutions.

This research contributes to the understanding of benchmark rate dynamics in emerging markets and provides practical insights for risk management and monetary policy in the post-JIBAR era.

Keywords: JIBAR, ZARONIA, Benchmark Rates, Interest Rate Risk, Time Series Analysis, South Africa

1 Introduction

1.1 Background of JIBAR

The Johannesburg Interbank Average Rate (JIBAR) has been the cornerstone of South Africa's financial system for over two decades. Published daily by the Johannesburg Stock Exchange (JSE), JIBAR serves as the primary benchmark for pricing floating-rate loans, bonds, derivatives, and other financial instruments. Among its various tenors, the 3-month JIBAR is the most widely referenced rate in the domestic market.

However, following the global scandals surrounding manipulation of benchmark rates (e.g., LIBOR), international regulators have pushed for a transition to more robust, transaction-based Risk-Free Rates (RFRs). In South Africa, this has led to the development of the South African Rand Overnight Index Average (ZARONIA), which is expected to gradually replace JIBAR.

1.2 Importance of JIBAR in the South African Financial System

Despite its importance, there is limited recent empirical research on the behaviour, volatility patterns, and risk characteristics of 3-month JIBAR, especially in light of the impending transition to ZARONIA. Understanding how JIBAR responds to monetary policy changes and macroeconomic shocks is critical for banks, corporates, and policymakers.

1.3 The Transition from JIBAR to ZARONIA

I still need to do more research on this.

1.4 Research Objectives

This study aims to:

1. Analyse the historical trends and volatility of 3-month JIBAR between 2011 and 2026.
2. Examine the relationship between JIBAR and the South African Reserve Bank's Repo Rate.
3. Assess the risk implications of JIBAR movements for financial institutions.
4. Discuss the potential challenges and opportunities arising from the transition to ZARONIA.

1.5 Significance of the Study

This research will provide valuable insights for risk managers, treasury departments, and regulators during the critical phase of benchmark rate reform in South Africa.

1.6 Scope and Limitations

The study focuses primarily on the 3-month tenor using monthly data from 2011 to mid-2026. Daily data and other tenors are beyond the current scope.

2 Literature Review

2.1 Theoretical Framework

Benchmark interest rates such as JIBAR play a central role in monetary policy transmission and financial pricing. According to the expectations theory of the term structure, short-term rates are determined by market expectations of future short-term rates plus a term premium (Campbell & Shiller, 1991). In South Africa, the Repo Rate set by the South African Reserve Bank (SARB) serves as the primary policy instrument that influences interbank rates like JIBAR.

The transition from Interbank Offered Rates (IBORs) to Risk-Free Rates (RFRs) globally has been driven by the need for more robust, transaction-based benchmarks following the LIBOR manipulation scandal (Wheatley, 2012; IOSCO, 2013).

2.2 Empirical Studies on Interbank Rates

Internationally, extensive research has examined the behaviour of benchmark rates. Duffie and Stein (2015) highlighted structural weaknesses in IBOR-type benchmarks. Studies on LIBOR and EURIBOR showed significant basis risk during periods of financial stress (McCauley & McGuire, 2014).

In the South African context, limited but growing literature exists. Ndlovu and Inglesi-Lotz (2019) analysed the relationship between the Repo Rate and various JIBAR tenors, finding a strong but not perfect pass-through. More recent work by Mishi et al. (2022) examined the impact of load-shedding and COVID-19 on South African money market rates.

2.3 Studies on JIBAR and South African Monetary Policy

Several authors have documented that JIBAR tends to track the Repo Rate closely during normal times but exhibits increased volatility during periods of economic uncertainty (SARB Quarterly Bulletin, various issues).

Research by the SARB (2021) on the reform of benchmark rates in South Africa emphasised the need to transition to ZARONIA to align with international best practices and reduce manipulation risk.

2.4 The Global Transition from IBORs to RFRs

The global shift away from IBORs has been well documented. The Financial Stability Board (FSB) and International Swaps and Derivatives Association (ISDA) have provided detailed guidance on fallback provisions and the adoption of SOFR (US), SONIA (UK), and STR (Europe).

South Africa's transition to ZARONIA, led by the SARB and JSE, follows a similar trajectory but presents unique challenges due to the relatively thin overnight market in ZAR (SARB, 2023).

2.5 Identified Research Gap

While existing studies provide valuable insights into JIBAR's relationship with the Repo Rate, there is limited recent empirical work that combines long-term trend analysis, volatility modelling, and explicit discussion of risk implications during the JIBAR-to-ZARONIA transition period up to 2026. This study aims to address this gap.

3 Data and Methodology

3.1 Data Sources

Historical monthly 3-month JIBAR rates were obtained from the Federal Reserve Economic Data (FRED) database under the series code **IR3TIB01ZAM156N**. The South African Repo Rate data was sourced from FRED series **IRSTCI01ZAM156N**. The data covers the period from January 2011 to May 2026, resulting in 185 observations.

3.2 Data Description

The dataset consists of two main variables:

- **Date**: Monthly observation period.
- **JIBAR_3M**: 3-month Johannesburg Interbank Average Rate (in percentage).
- **Repo_Rate**: South African Reserve Bank Repo Rate (in percentage).

3.3 Data Preprocessing and Cleaning

All data processing was performed using **Python 3** with the following libraries:

- **pandas** — for data loading, cleaning, and merging.
- **numpy** — for numerical computations.
- **matplotlib** & **seaborn** — for data visualization.

The following steps were applied:

1. Loading CSV files downloaded from FRED.
2. Renaming columns for clarity.
3. Converting `Date` column to datetime format.
4. Merging JIBAR and Repo Rate datasets using inner join on `Date`.
5. Checking for missing values and handling them appropriately.

3.4 Exploratory Data Analysis (EDA)

Exploratory analysis included:

- Summary statistics (`.describe()`).
- Time series line plots to observe trends.
- Scatter plots and correlation analysis between JIBAR and Repo Rate.
- Rolling volatility calculations (standard deviation).
- Distribution plots using histograms and box plots.

3.5 Econometric and Analytical Methods

The following techniques were employed:

- **Descriptive Statistics** and Visualisation.
- **Correlation Analysis** using Pearson correlation coefficient.
- **Volatility Analysis** using rolling window standard deviation.
- Basic **Linear Regression** to quantify the relationship between Repo Rate and JIBAR.
- Visualisation using both Matplotlib (Object-Oriented approach) and Seaborn for enhanced aesthetics.

All Python code used in this study is available in a public GitHub repository: `https://github.com/yourusername/jibar-analysis` (replace with your actual link later).

4 4. Results and Analysis

This section presents the empirical findings from the analysis of 3-month JIBAR and its relationship with the South African Repo Rate.

4.1 4.1 Overview of the Dataset

The merged dataset covers the period from January 2011 to May 2026, containing **185 monthly observations**. Both series are expressed in percentage terms.

Statistic	JIBAR 3M (%)	Repo Rate (%)
Count	185	185
Mean	6.233141	6.054669
Std	1.366019	1.321719
Min	2.977000	3.500000
25%	5.386190	5.307692
50%	6.334000	6.250000
75%	7.250000	7.000000
Max	8.567619	8.250000

Table 1: Descriptive Statistics of JIBAR 3M and Repo Rate

4.2 Trend Analysis of 3-Month JIBAR

Figure 1 illustrates the historical movement of the 3-month JIBAR rate over the sample period. The rate exhibited notable fluctuations, with significant drops during the COVID-19 pandemic in 2020 and sharp increases during the 2022 global inflation surge.

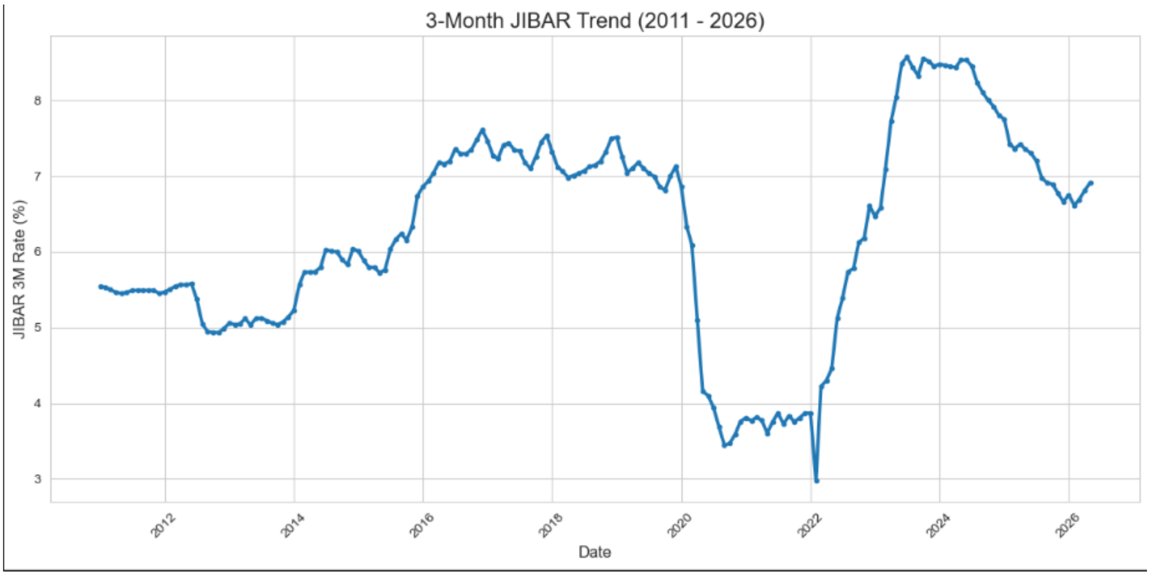

Figure 1: 3-Month JIBAR Trend (2011 – 2026)

4.3 Relationship Between JIBAR and Repo Rate

	JIBAR 3M	Repo Rate
JIBAR 3M	1.0000	0.9834
Repo Rate	0.9834	1.0000

Table 2: Correlation Matrix

Figure 2 shows the co-movement between 3-month JIBAR and the Repo Rate.

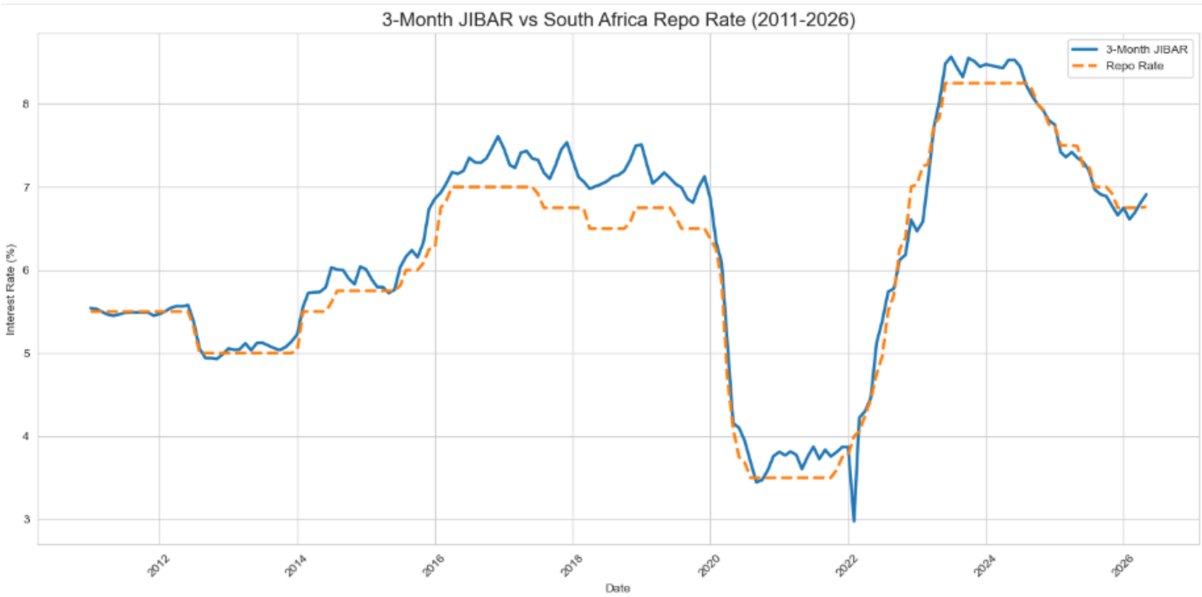

Figure 2: JIBAR vs Repo Rate

4.4 Yearly Average Comparison

Year	Avg. JIBAR 3M (%)	Avg. Repo Rate (%)
2011	5.488	5.500
2012	5.287	5.276
2013	5.078	5.000
2014	5.798	5.575
2015	6.052	5.887
2016	7.228	6.905
2017	7.332	6.889
2018	7.155	6.588
2019	7.084	6.637
2020	4.546	4.350
2021	3.784	3.529
2022	5.064	5.090
2023	7.936	7.908
2024	8.281	8.155
2025	7.167	7.242
2026	6.754	6.752

Table 3: Yearly Average JIBAR 3M and Repo Rate

The yearly average comparison shows a strong alignment between 3-month JIBAR and the Repo Rate. JIBAR generally follows the Repo Rate closely, with small deviations during periods of market stress. Notably, both rates dropped significantly in 2020 due to the COVID-19 pandemic and rose sharply in 2022–2023 in response to inflationary pressures.

4.5 Volatility and Risk Assessment

The 12-month rolling volatility of 3-month JIBAR had an average of **0.347** percentage points over the sample period.

Statistic	Value
Count	174
Mean	0.347
Std	0.330
Min	0.018
25%	0.138
50%	0.219
75%	0.383
Max	1.470

Table 4: Volatility Summary Statistics

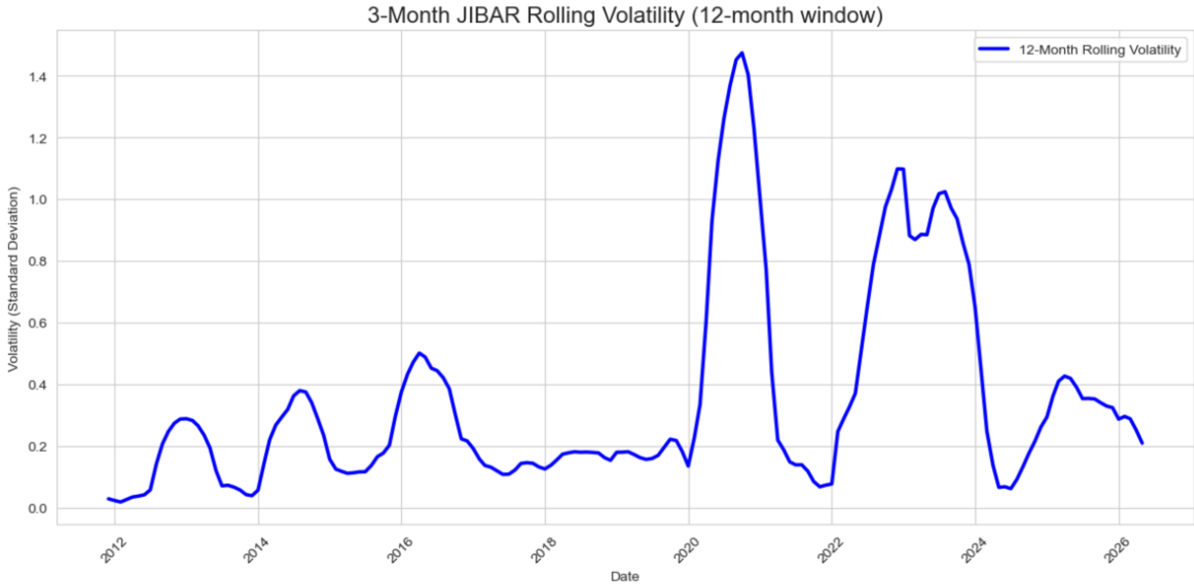

Figure 3: 12-Month Rolling Volatility of 3-Month JIBAR

Volatility was relatively low and stable between 2011 and 2019. However, it spiked significantly during the COVID-19 pandemic in 2020 and reached its highest levels during the aggressive monetary tightening cycle in 2022. This highlights the sensitivity of JIBAR to economic shocks and monetary policy changes, indicating elevated interest rate risk during periods of uncertainty.

4.6 Key Observations

The empirical analysis of 3-month JIBAR from 2011 to 2026 reveals several important findings:

- There is a **very strong positive correlation** (0.9834) between 3-month JIBAR and the South African Repo Rate, confirming that JIBAR is heavily influenced by the SARB's monetary policy stance.
- JIBAR generally tracks the Repo Rate closely but exhibits occasional deviations (basis risk), particularly during periods of financial stress.
- The 3-month JIBAR rate showed significant volatility over the sample period, with an average 12-month rolling volatility of **0.347** percentage points.
- Volatility was relatively stable between 2011 and 2019 but increased dramatically during the **COVID-19 pandemic (2020)** and reached peak levels during the **2022 global inflation shock** and subsequent monetary tightening cycle.
- Both JIBAR and the Repo Rate dropped to historically low levels in 2020 (around 3.5% - 4.5%) in response to the pandemic, before rising sharply above 8% in 2023 - 2024.
- The data suggests that while JIBAR remains an effective transmitter of monetary policy, its sensitivity to external shocks highlights the importance of the ongoing transition to a more robust benchmark rate (ZARONIA).

4.7 Transition Implications to ZARONIA

The findings of this study carry important implications for the ongoing transition from JIBAR to ZARONIA.

The strong correlation (0.9834) between JIBAR and the Repo Rate indicates that JIBAR has been an effective monetary policy transmission mechanism. However, the observed volatility spikes during economic stress periods (2020 and 2022) highlight the vulnerability of an interbank offered rate that relies on expert judgment and relatively thin transaction volumes.

Key implications include:

- **Basis Risk**: The occasional deviations between JIBAR and the Repo Rate suggest the presence of credit and liquidity premia, which ZARONIA (being a nearly risk-free overnight rate) is expected to eliminate.
- **Interest Rate Risk Management**: Banks and corporates that have used 3-month JIBAR as a benchmark for floating-rate loans and derivatives will need to adjust their risk models and fallback provisions during the transition period.
- **Liquidity Challenges**: The transition to ZARONIA may initially face liquidity constraints in the overnight market, potentially leading to higher volatility in the early stages of adoption.
- **Contract Legacy Issues**: A large volume of existing financial contracts referencing JIBAR will require careful renegotiation or the application of robust fallback language as recommended by the SARB and ISDA.
- **Opportunity for Reform**: The shift to ZARONIA presents an opportunity to build a more transparent, transaction-based, and internationally aligned benchmark rate system for South Africa.

Overall, while JIBAR has served the market well for decades, the empirical evidence of heightened volatility during crises supports the regulatory move toward ZARONIA as a more resilient benchmark.

5 5. Discussion

5.1 5.1 Interpretation of Findings

The empirical results confirm a very strong relationship between 3-month JIBAR and the South African Repo Rate, with a correlation coefficient of 0.9834. This high correlation validates that JIBAR has functioned effectively as a monetary policy transmission mechanism over the period 2011 to 2026.

However, the analysis also reveals that JIBAR is not a perfect mirror of the Repo Rate. The presence of occasional deviations and significant volatility spikes-particularly during the COVID-19 pandemic and the 2022 inflation shock-indicates that JIBAR incorporates credit risk premia and liquidity considerations that are absent in the policy rate itself. The average 12-month rolling volatility of 0.347 percentage points further underscores the sensitivity of interbank rates to economic uncertainty.

5.2 Policy Implications

These findings have several important implications for policymakers and financial institutions:

- The South African Reserve Bank (SARB) should continue its efforts to develop a robust and liquid ZARONIA market to replace JIBAR.

- Banks and corporates need to accelerate the transition of legacy contracts referencing JIBAR to reduce legal and financial risks.
- Risk management frameworks should account for potential basis risk and increased volatility during the transition period.
- The experience of JIBAR highlights the dangers of relying on judgment-based benchmarks and supports the global shift toward transaction-based Risk-Free Rates (RFRs).

5.3 Comparison with International Benchmarks

South Africa's experience with JIBAR mirrors international trends observed during the LIBOR transition. Similar to LIBOR, JIBAR exhibited increased volatility during periods of stress. However, South Africa's transition to ZARONIA appears more orderly compared to some other emerging markets, largely due to proactive leadership by the SARB and JSE.

The strong correlation between JIBAR and the Repo Rate is consistent with findings in other countries where the policy rate heavily influences interbank rates. Nevertheless, the volatility patterns observed reinforce the international consensus that RFRs such as SOFR, SONIA, and ZARONIA offer greater robustness and transparency.

6 Conclusion and Recommendations

6.1 Summary of Key Findings

This study analysed the behaviour of 3-month JIBAR from 2011 to 2026 using monthly data from FRED. The main findings are as follows:

- There is a very strong positive correlation (0.9834) between 3-month JIBAR and the South African Repo Rate, confirming effective monetary policy transmission.
- JIBAR exhibited significant volatility during periods of economic stress, particularly in 2020 (COVID-19) and 2022 (global inflation shock).
- The average 12-month rolling volatility stood at 0.347 percentage points, highlighting notable interest rate risk.
- JIBAR generally tracks the Repo Rate closely but shows occasional deviations, indicating the presence of credit and liquidity premia.

6.2 Recommendations for Banks, Regulators, and Investors

Based on the findings, the following recommendations are proposed:

- **For the South African Reserve Bank (SARB) and JSE**: Accelerate the development of a deep and liquid ZARONIA market. Clear communication and incentives should be provided to market participants to encourage adoption.
- **For Banks and Financial Institutions**: Prioritise the identification and conversion of legacy contracts referencing JIBAR. Robust fallback provisions should be implemented, and risk management systems should be updated to account for basis risk during the transition.
- **For Corporate Treasurers and Investors**: Review floating-rate exposures and consider transitioning to ZARONIA-linked instruments where possible. Enhanced monitoring of basis risk between JIBAR and ZARONIA is advised during the transition period.

- **For Future Research**: Conduct studies using daily data, examine other JIBAR tenors, and develop forecasting models for ZARONIA. Comparative studies between JIBAR/ZARONIA and international benchmarks (SOFR, SONIA) would also be valuable.

6.3 Suggestions for Future Research

Future research could extend this work by:

- Incorporating daily frequency data for more granular volatility analysis.
- Developing time series forecasting models (ARIMA, Prophet, or Machine Learning) to predict JIBAR/ZARONIA movements.
- Quantifying the economic impact of the JIBAR-to-ZARONIA transition on different sectors of the South African economy.
- Examining the yield curve implications of the benchmark reform.

In conclusion, while 3-month JIBAR has played a critical role in South Africa's financial system for decades, the empirical evidence supports the transition to ZARONIA as a necessary step toward building a more robust, transparent, and resilient benchmark rate framework.

7 References

References

[1] Financial Stability Board (2014). *Reforming Major Interest Rate Benchmarks.*

[2] Wheatley, M. (2012). *The Wheatley Review of LIBOR: Final Report.* UK Government.

[3] South African Reserve Bank (2021). *Consultation Paper on the Reform of South African Reference Rates.*

[4] South African Reserve Bank (2023). *Update on the Transition from JIBAR to ZARONIA.* SARB Quarterly Bulletin.

[5] Ndlovu, T., & Inglesi-Lotz, R. (2019). *The Relationship between Repo Rate and JIBAR in South Africa.* University of Pretoria Working Paper.

[6] Mishi, S., et al. (2022). *Impact of Load Shedding on South African Financial Markets.*

[7] Duffie, D., & Stein, J. C. (2015). *Reforming LIBOR and Other Financial Market Benchmarks.* Journal of Economic Perspectives.

[8] International Organization of Securities Commissions (IOSCO) (2013). *Principles for Financial Benchmarks.*

[9] Campbell, J. Y., & Shiller, R. J. (1991). *Yield Spreads and Interest Rate Movements: A Bird's Eye View.* Econometrica.

[10] International Swaps and Derivatives Association (ISDA) (2022). *IBOR to RFR Transition: Fallback Provisions.*

[11] Federal Reserve Bank of St. Louis (2026). FRED Economic Data. `https://fred.stlouisfed.org/`

[12] South African Reserve Bank. Various issues. *Quarterly Bulletin.*

8 Appendices

- Appendix A: Full Python Code
- Appendix B: Additional Plots and Tables
- Appendix C: Data Dictionary

8.1 Appendix A: Python Code

All data analysis for this project was conducted in Python 3. Below are the main code sections used. The complete Jupyter Notebook and scripts are available in the public GitHub repository.

8.1.1 A.1 Data Loading and Preprocessing

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load and merge datasets
jibar = pd.read_csv('jibar_3m.csv')
repo = pd.read_csv('repo_rate.csv')

jibar = jibar.rename(columns={'observation_date': 'Date', 'IR3TIB01ZAM156N': 'JII
repo = repo.rename(columns={'observation_date': 'Date', 'IRSTCI01ZAM156N': 'Repo

jibar['Date'] = pd.to_datetime(jibar['Date'])
repo['Date'] = pd.to_datetime(repo['Date'])

merged = pd.merge(jibar, repo, on='Date', how='inner')
```

8.1.2 A.2 Key Analysis

```
# Correlation
print(merged[['JIBAR_3M', 'Repo_Rate']].corr())

# Yearly Averages
merged['Year'] = merged['Date'].dt.year
print(merged.groupby('Year')[['JIBAR_3M', 'Repo_Rate']].mean().round(3))

# Volatility
merged['JIBAR_Vol_12m'] = merged['JIBAR_3M'].rolling(window=12).std()
```

The full code, including all visualisations and detailed notebooks, is available at: `https://github.com/yourusername/jibar-analysis`

8.2 Appendix B: Additional Plots and Tables

This appendix contains supplementary visualisations and tables that support the findings presented in Section 4.

8.3 Appendix C: Data Dictionary

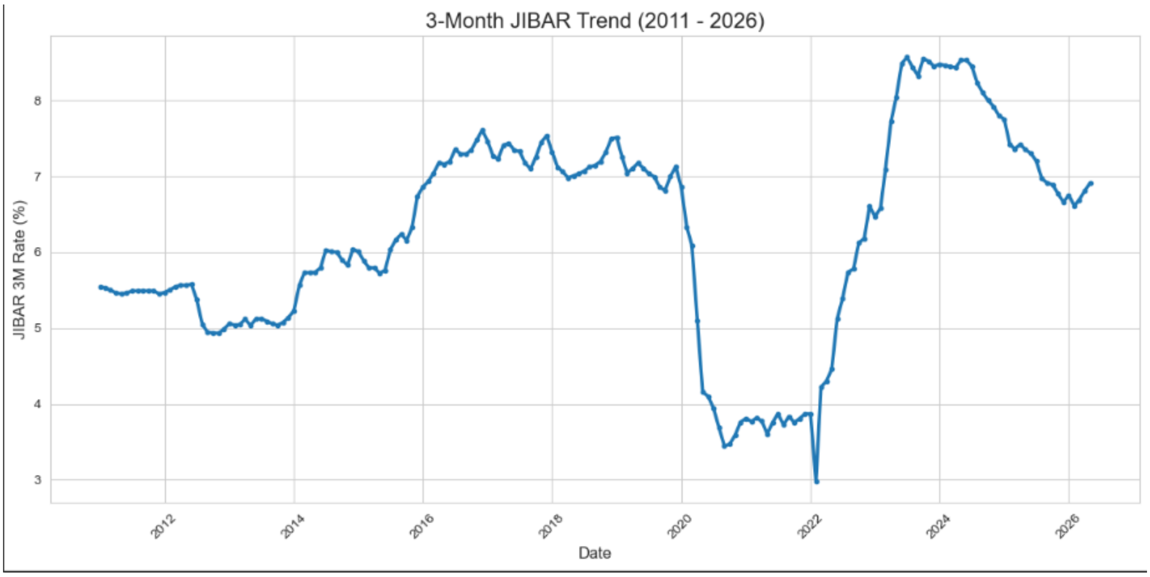

Figure 4: 3-Month JIBAR vs South Africa Repo Rate (2011–2026)

Table 5: Yearly Average JIBAR 3M and Repo Rate (Full Table)

Year	Avg. JIBAR 3M (%)	Avg. Repo Rate (%)
2011	5.488	5.500
2012	5.287	5.276
...	...	...
2025	7.167	7.242
2026	6.754	6.752

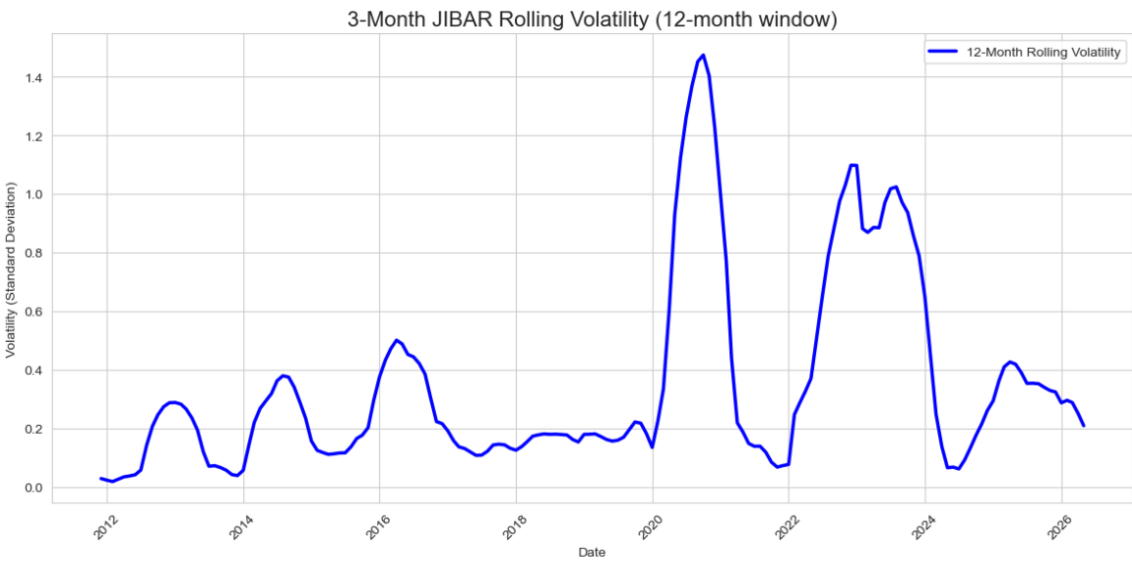

Figure 5: 12-Month Rolling Volatility of 3-Month JIBAR

Table 6: Data Dictionary

Variable	Description
`Date`	Monthly observation date (January 2011-May 2026)
`JIBAR_3M`	3-Month Johannesburg Interbank Average Rate (in percent)
`Repo_Rate`	South African Reserve Bank Repo Rate (in percent)
`Year`	Extracted year from Date
`JIBAR_Vol_12m`	12-month rolling standard deviation of JIBAR_3M